

1 Pre-training

Gemini is powered by Google’s most capable AI models, designed with varying capabilities and use cases. Like most LLMs today, these models are pre-trained on a variety of data from publicly available sources. We apply quality filters to all datasets, using both

2 Post-training

After the initial training, LLMs go through additional steps to refine their responses. One of these is called Supervised Fine-Tuning (SFT), which trains the model on carefully selected examples of excellent answers. It's like teaching children to write by showing them well-written stories and essays.

Next comes Reinforcement Learning from Human Feedback (RLHF). Here, the model learns to generate even better responses based on scores or feedback from a special Reward Model. This Reward Model is trained on human preference data, where responses have been rated relative to one another, teaching it what people prefer. Preference data may sometimes include and expose models to offensive or incorrect data so that they learn how to recognize it and avoid it. You can think of preference data like rewarding a child for a job well done; the model is rewarded for creating answers that people like.